

JavaScript

What Is JavaScript in a Website?

JavaScript is the part of a website that controls how the page behaves. It decides what happens when time passes or when people interact with the page.

- HTML creates the content
- CSS controls the look
- JavaScript controls actions
- JavaScript makes pages respond
- Without it, pages feel inactive

Why Websites Need JavaScript

A page can look complete but still feel “dead” if it cannot react to people. JavaScript gives the page the ability to respond.

- Buttons need JavaScript
- Messages need JavaScript
- Changes need JavaScript
- Interaction needs JavaScript
- Real websites rely on it

JavaScript as Behavior

JavaScript controls behavior, not appearance or structure. Behavior means how the page changes over time or reacts to actions.

- Behavior = what happens
- Not how it looks
- Not what exists
- How the page reacts
- How the page changes

Ways a Page Can Behave

A web page can behave in different ways depending on time and user interaction. JavaScript handles all of them.

- Reacting to clicks
- Reacting to typing
- Reacting to scrolling
- Reacting to time
- Reacting to page load

Time-Based Behavior

JavaScript can make things happen automatically after a certain time or when the page loads.

- Change text on load
- Show messages later
- Update content
- Control timing
- No user action needed

User Interaction Behavior

JavaScript listens to users and reacts when they do something on the page.

- Clicking buttons
- Typing in inputs
- Hovering elements
- Scrolling pages
- Submitting forms

Cause and Effect on Pages

JavaScript works using a simple rule: when something happens, something changes on the page.

- User action = cause
- Page change = effect
- Click → change text
- Click → show message
- Simple reaction logic

How JavaScript Connects to HTML

JavaScript must be connected to HTML so it can see and control the page. Without this connection, JavaScript cannot work.

- Linked using `<script>`
- HTML loads first
- JavaScript runs after
- Wrong place breaks behavior
- Correct place enables behavior

Why HTML Must Exist First

JavaScript cannot control elements that do not exist yet. The page must be built before JavaScript can interact with it.

- No HTML = no control
- JavaScript needs elements
- Page loads first
- Behavior comes after
- Structure before behavior

Selecting Elements on the Page

JavaScript must point to specific elements before it can change them.
Selecting means choosing one thing on the page.

- Select a button
- Select a title
- Select a message
- One element at a time
- Similar to CSS selectors

What JavaScript Can Change

Once JavaScript selects an element, it can change many visible things on the page.

- Text content
- Images
- Colors
- Visibility
- CSS classes

Visibility Changes

JavaScript often controls whether something is visible or hidden on the page.

- Show messages
- Hide messages
- Toggle visibility
- Change user feedback
- Improve interaction

Page State (Modes)

A page can have different modes depending on what the user has done. JavaScript controls which mode is active.

- Before interaction
- After interaction
- Active mode
- Inactive mode
- Visible state changes

State Is Visible

State is not abstract. You can always see it on the page through changes in text, color, or layout.

- Button text changes
- Message appears
- Style changes
- Layout updates
- Visual confirmation

Learning by Breaking Code

Making mistakes in JavaScript is normal. Errors help us understand how the page and behavior work together.

- Errors are common
- Errors are safe
- Errors show limits
- Errors teach structure
- Debugging builds confidence

Fixing Broken Behavior

When behavior stops working, the solution is usually simple and visible in the code or connection.

- Check element selection
- Check script location
- Check spelling
- Check logic flow
- Small fixes matter

Building Pages with JavaScript

JavaScript can create page elements by itself, but this is not always the best starting point for beginners.

- JS can create elements
- JS can remove elements
- JS can rebuild pages
- More complex approach
- Less visual at first

Adding Behavior to Existing Pages

The most common and beginner-friendly approach is to build the page first, then add JavaScript behavior.

- HTML first
- CSS second
- JavaScript last
- Clear separation
- Easier to understand

When to Use Each Approach

Knowing the difference helps you choose the right method for your project and skill level.

- Static page + behavior = simple
- Full JS build = advanced
- Most websites mix both
- Start simple
- Grow complexity later

For any questions please ask on
Discord server technical questions channel

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